Representation Learning with Autoencoders (AE & VAE)

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1. Model Architectures

1.1 Autoencoder (AE)

The Autoencoder consists of an encoder–decoder architecture. The encoder compresses input images (64×64 grayscale) into a lower-dimensional representation, while the decoder reconstructs the original image from this representation.

- **Encoder**:
  - Conv2D (16 filters) + MaxPooling
  - Conv2D (32 filters) + MaxPooling
  - Output: compressed feature map
- **Decoder**:
  - Conv2DTranspose (32 filters)
  - Conv2DTranspose (16 filters)
  - Final Conv2D (sigmoid activation)

1.2 Variational Autoencoder (VAE)

The VAE extends the AE by introducing a probabilistic latent space.

- The encoder outputs:
  - **z_mean**
  - **z_log_var**
- The decoder reconstructs images from sampled latent vectors.

The VAE loss consists of:

- **Reconstruction loss** (binary cross-entropy)
- **KL divergence loss**

2. Key Differences Between AE and VAE

Aspect	AutoEncoder(AE)	Variational Autoencoder (VAE)
Latent space	Deterministic	Probabilistic
Output quality	Sharper	Smoother (slightly blurred)
Generation ability	None	Can generate new samples
Loss function	Reconstruction loss	Reconstruction +KL divergence

3. Latent Space Behavior Analysis

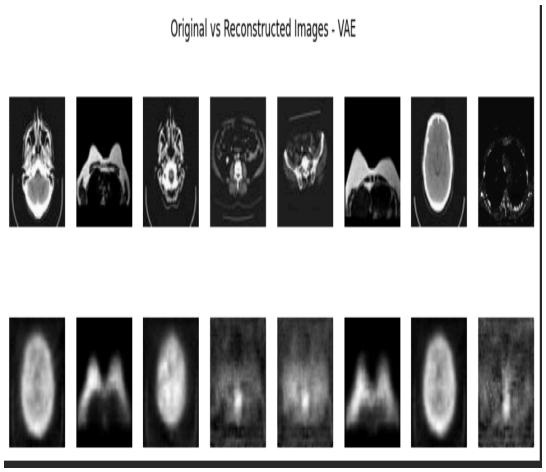
The latent space of the VAE was visualized in 2D using the learned mean vectors. The plot shows partial clustering of different medical image classes (AbdomenCT, BreastMRI, CXR, ChestCT, Hand, HeadCT).

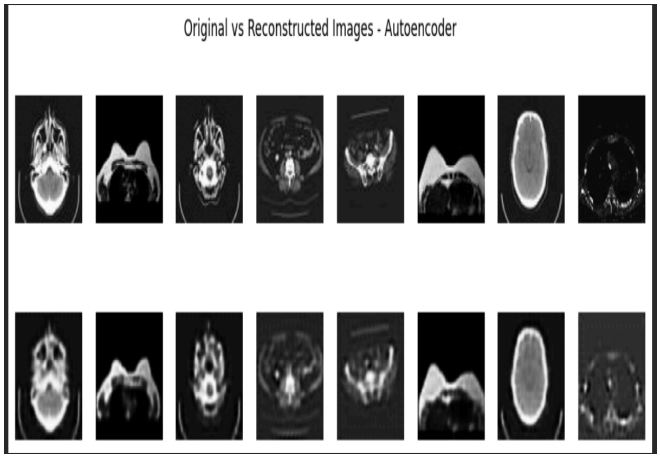
Some classes overlap due to visual similarity (e.g., CT scans), while others form more distinct clusters. This indicates that the VAE successfully learned a meaningful low-dimensional representation, although separation is not perfect due to similarities in medical imaging patterns.

4. Results and Insights

4.1 Reconstruction Performance

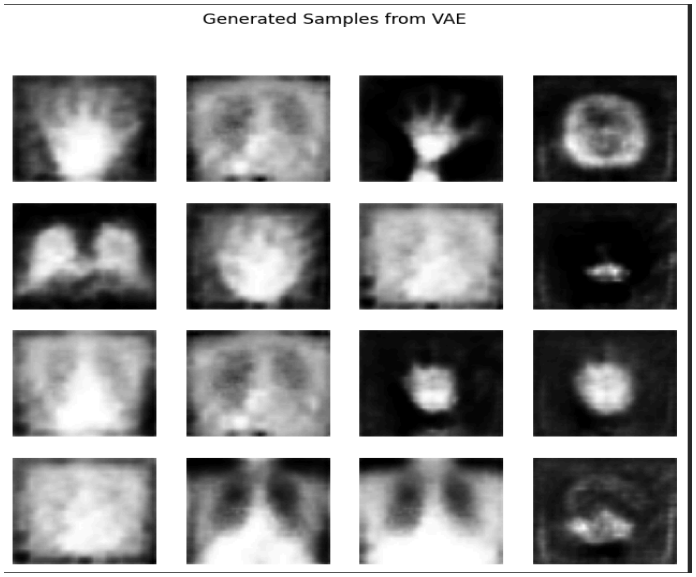
The Autoencoder achieved lower reconstruction error (MSE and MAE) compared to the VAE(The AE achieved MSE = 0.00235 and MAE = 0.02613, while the VAE achieved MSE = 0.02044 and MAE = 0.08646.), producing sharper and more detailed reconstructions. In contrast, the VAE outputs were smoother and slightly blurred due to the stochastic sampling process.





4.2 Sample Generation

The VAE successfully generated new medical-like images by sampling from the latent space. This demonstrates its generative capability, which is not present in standard autoencoders.



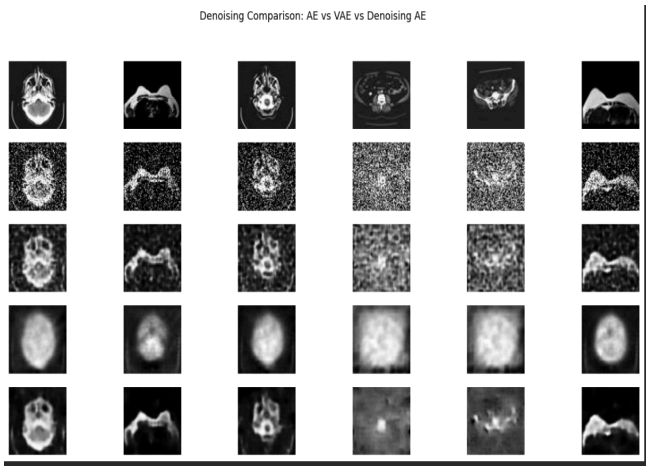
4.3 Denoising Capability

To evaluate robustness, Gaussian noise was added to input images. Three models were compared:

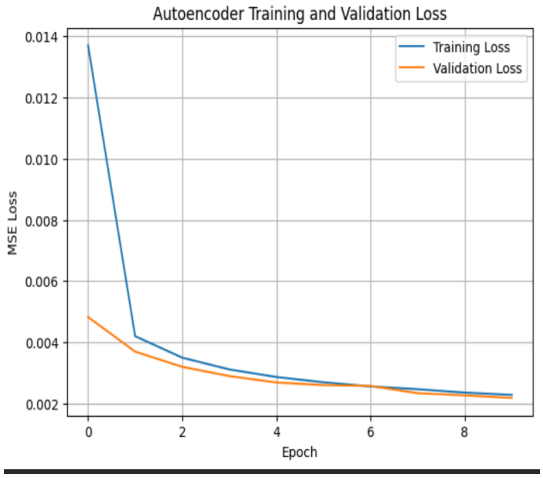
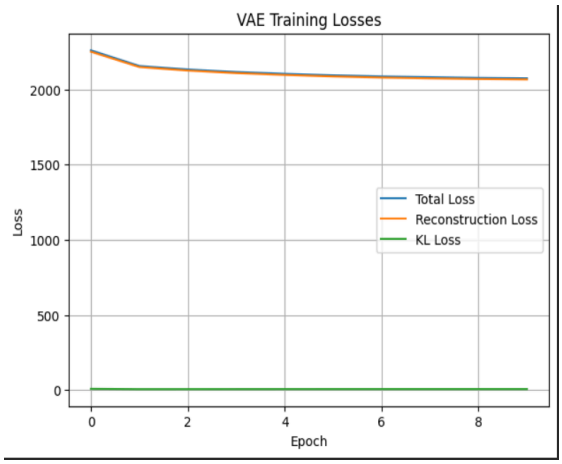
- AE (trained on clean images)
- VAE (trained on clean images)
- Denoising Autoencoder (trained on noisy images)

Results show:

- AE removes some noise but retains artifacts
- VAE produces smoother but blurrier outputs
- Denoising AE achieves the best performance, effectively removing noise while preserving structure



4.4 Loss Analysis

Training loss curves show that both AE and VAE converge successfully. For VAE:

- Reconstruction loss decreases steadily
- KL divergence stabilizes
- Total loss converges, indicating proper balance between reconstruction and regularization